

ScanTool.net Forum

For Software Developers => Developer Forum => Topic started by: STN-Brian on April 17, 2018, 09:48:17 am

Title: **Intro to OBD development**

Post by: **STN-Brian** on **April 17, 2018, 09:48:17 am**

Below are some links to resources that may help you get started in your OBD development:

API:

The FRPM (Family Reference and Programming Manual) describes the architecture, features, and the command set of the STN family of OBD interpreters.

- OBDLink FRPM (<https://www.obdlink.com/frpm>)

Datasheets:

The datasheets contain information such as pinouts, guidelines, and references schematics

- STN1110 datasheet (https://www.scantool.net/scantool/downloads/313/stn1110_ds_f.pdf)
- STN1170 datasheet (<https://www.scantool.net/scantool/downloads/132/stn1170-dse.pdf>)
- STN21xx datasheet (<https://www.scantool.net/scantool/downloads/206/stn21xx-dsac.pdf>)

OBD-II official standard:

For more complete information about OBD-II, we recommend the 5 standards below. These include most of what you need to know to get started, including protocol specific header information, PIDs, and DTCs. J1979, J1979-2, and J1979-3 and J1979DA will define service modes and their PIDs, while J2012 defines DTCs.

- SAE J1979 (https://www.sae.org/standards/content/j1979_201702/)
- SAE J1979-2 (OBDonUDS) (https://www.sae.org/standards/content/j1979-2_202104/)
- SAE J1979-3 (ZEVonUDS) (https://www.sae.org/standards/content/j1979-3_202310/)
- SAE J1979DA (https://www.sae.org/standards/content/j1979da_202404/)
- SAE J2012 (https://www.sae.org/standards/content/j2012_201612/)

Wikipedia also provides some information about OBD-II PIDs and the pinout for the DLC (OBD-II port)

- Wikipedia - OBD-II PIDs (https://en.wikipedia.org/wiki/OBD-II_PIDs)
- Wikipedia - DLC (https://en.wikipedia.org/wiki/On-board_diagnostics#OBD-II_diagnostic_connector)

Protocol Detection:

Below are the some of the basic steps to follow in order to automatically detect the protocol and start requesting PIDs.

- Check if there has been a power cycle event (**STDICPO**) to determine if the scan tool has potentially changed vehicles
- If there was, you should set protocol detection to automatic (**ATSP 0**)
- Turn on headers to see which ECUs are responding (**ATH 1**)
- Request supported PIDs (**0100**)
 - 0100 response can be decoded using the J1979DA standard or Wikipedia (https://en.wikipedia.org/wiki/OBD-II_PIDs#Service_01_PID_00)
- Determine which protocol was detected (**STPR** or **STPRS**)
- Hard set the protocol using the **STP** command

- Request desired PID from list of supported PIDs

Function and physical addressing:

Below is a link to a post that discussed functional and physical addressing:

Physical and functional ECU headers (<https://www.scantool.net/forum/index.php?topic=11518>)

OBLink Knowledge base:

The OBLink Knowledge base (<https://support.obdlink.com/support/solutions>) is a comprehensive resource hub designed to assist users with a variety of needs related to OBLink products. This ever-expanding repository of articles covers a wide range of topics including:

- **Getting Started:** Guides and tutorials for new users to help them set up and begin using their OBLink devices efficiently.
- **OBLink App Help:** Detailed instructions and tips for navigating and making the most of the OBLink app features.
- **Troubleshooting:** Solutions and step-by-step guides to address common issues and technical problems encountered by users.
- **Advanced Topics:** In-depth articles for more experienced users looking to explore advanced functionalities and integrations.
- **Product Information:** Specifications, features, and usage tips for all OBLink products.

Whether you're a new user looking for setup assistance or an experienced user needing troubleshooting tips, the OBLink Knowledge Base provides valuable information to enhance your experience with OBLink products.